

Redox Reactions

* Redox reactions are the reactions in which **oxidation** and **reduction** occur simultaneously.

OIL
RIG

1 Key Terms.

- **Oxidation** : Addition of oxygen / Removal of hydrogen / Loss of e^- / Increase in oxidation number.
- **Reduction** : Addition of hydrogen / Removal of oxygen / Gain of e^- / Decrease in oxidation number.
- **Oxidising agent** : Species that oxidises others (it gets reduced).
- **Reducing agent** : Species that reduces others (it gets oxidised).

OIL RIG Concept

Oxidation is
Loss of e^-
Reduction is
Increase of e^-
Gain / loss in
oxidation number



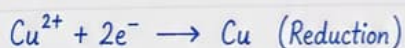
2 Rules for Assigning Oxidation Number (O.N.).

- O.N. of free element = 0
e.g., O_2 , Na, Fe
- O.N. of monatomic ion = charge on ion
e.g., $Na^+ = +1$, $Cl^- = -1$
- O.N. of O = -2 (except in peroxides where O = -1, in OF_2 where O = +2)
- O.N. of H = +1 (except in metal hydrides where H = -1)
- Sum of O.N. in a neutral molecule = 0
Sum of O.N. in a polyatomic ion = charge on ion

3 Example of Redox Reaction.

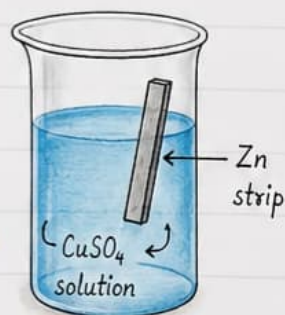


Zn is oxidised (Reducing agent)



Cu^{2+} is reduced (Oxidising agent)

Oxidation and reduction occur simultaneously.



Chapter - 7

Redox Reactions

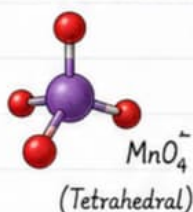
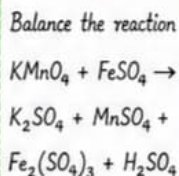
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CHEMISTRY

4 Balancing Redox Reactions

Steps (in acidic medium):

- Assign oxidation numbers to all atoms.
- Write the two half reactions (oxidation & reduction).
- Balance atoms other than H and O.
- Balance O by adding H_2O .
- Balance H by adding H^+ .
- Balance charge using e^- .
- Equalise electrons in both half reactions.
- Add the two half reactions and cancel common species.

Example



5 Oxidation vs Reduction

OXIDATION	REDUCTION
Addition of oxygen	Removal of oxygen
Removal of hydrogen	Addition of hydrogen
Loss of electrons (e^-)	Gain of electrons (e^-)
Increase in oxidation number	Decrease in oxidation number
Oxidising agent is reduced	Reducing agent is oxidised
Example : $2Cu + O_2 \rightarrow 2CuO$ (Cu is oxidised)	Example : $CuO + H_2 \rightarrow Cu + H_2O$ (Cu is reduced)

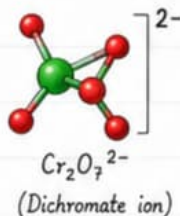
6 Oxidising and Reducing Agents

Oxidising Agent (Oxidant)

- Accepts electrons.
- Causes oxidation of other species.
- It is reduced itself.

e.g., $KMnO_4$, $K_2Cr_2O_7$, Cl_2

Oxidant gets reduced.

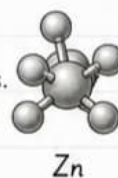


Reducing Agent (Reductant)

- Donates electrons.
- Causes reduction of other species.
- It is oxidised itself.

e.g., Zn, $FeSO_4$, SO_2

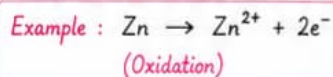
Reductant gets oxidised.



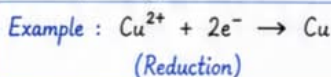
7 Half Reactions

- A redox reaction can be represented as a combination of two half reactions.
- Half reaction shows either oxidation or reduction.

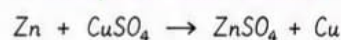
Oxidation Half Reaction

Species loses e^- .

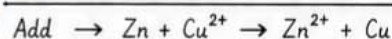
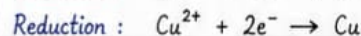
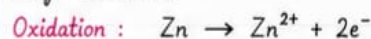
Reduction Half Reaction

Species gains e^- .

Example (Overall Reaction)

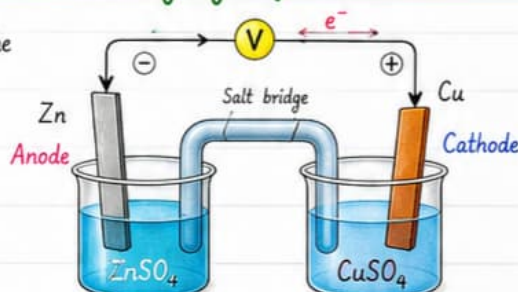


Half reactions :



8 Redox Reactions in Everyday Life - Electrochemical Cells

Redox reactions are the basis of working of galvanic (voltaic) cells such as Daniell cell.



In Daniell Cell:

- Oxidation occurs at anode (Zn)
- Reduction occurs at cathode (Cu)
- Electrons flow from anode to cathode
- Salt bridge maintains electrical neutrality

Key Takeaway

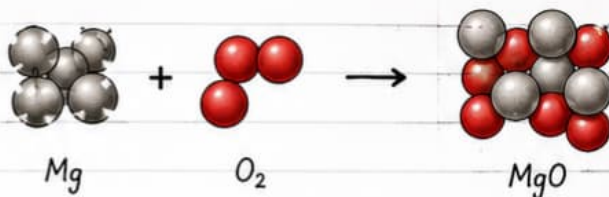
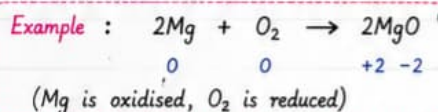
Redox reactions involve simultaneous oxidation and reduction. They are essential in nature, industries, and our daily lives.

- ✓ Energy production
- ✓ Corrosion
- ✓ Metallurgy
- ✓ Electrochemical cells

9 Types of Redox Reactions

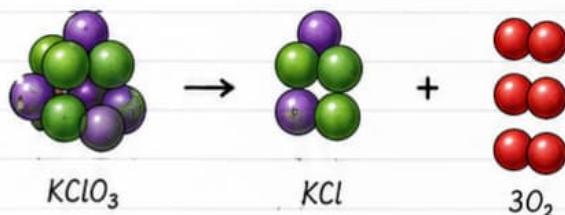
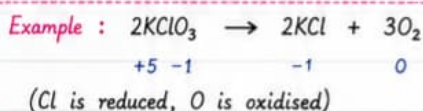
1. Combination Reaction (Synthesis)

Two or more substances combine and one is oxidised while the other is reduced.



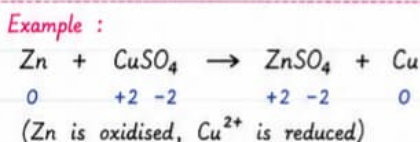
2. Decomposition Reaction

A compound breaks down and one element is oxidised while the other is reduced.



3. Displacement Reaction

More reactive element displaces less reactive element from its compound.



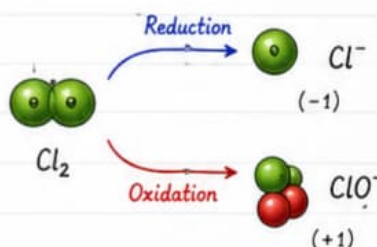
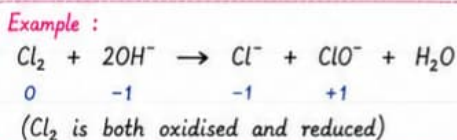
Reactivity Series

$\text{K} > \text{Na} > \text{Ca} > \text{Mg} > \text{Al} > \text{Zn} > \text{Fe} > \text{Pb} > (\text{H}) > \text{Cu} > \text{Hg} > \text{Ag} > \text{Au}$

More reactive metal displaces less reactive metal.

4. Disproportionation Reaction

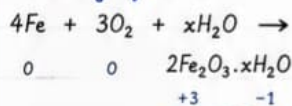
Same species gets oxidised as well as reduced simultaneously.



In disproportionation reaction, the oxidation number of the element increases in one part and decreases in another part.

10 Redox Reactions in Nature

Rusting of Iron



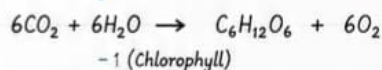
- Fe is oxidised to Fe^{3+} .
- O_2 is reduced.

Respiration in Living Beings



- Glucose is oxidised.
- O_2 is reduced.

Photosynthesis in Plants



- CO_2 is reduced.
- H_2O is oxidised.

11 Importance of Redox Reactions

Energy Production



Basis of battery, fuel cells and respiration.

Corrosion & Prevention



Helps in understanding rusting and methods to prevent it.

Metallurgy



Extraction of metals from their ores.

Industrial Applications



Used in bleaching, electroplating, extraction, etc.

Environmental Processes



Involved in nutrient cycles and pollution control.

Quick Revision

- Oxidation → addition of O / removal of H / loss of e^- / increase in O.N.
- Reduction → addition of H / removal of O / gain of e^- / decrease in O.N.
- Redox reactions → oxidation and reduction occur simultaneously.

Remember :

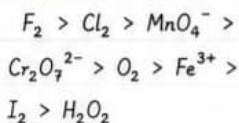
LEO the lion says
"Lose Electrons - Oxidation"
"Gain Electrons - Reduction"



12 Oxidising and Reducing Strength

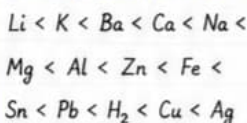
- Stronger oxidising agent → gets reduced easily.
- Stronger reducing agent → gets oxidised easily.

Oxidising Strength Order



Stronger oxidising agent →

Reducing Strength Order



Stronger reducing agent →

- ★ Stronger oxidising agent has higher tendency to accept e^- .
Stronger reducing agent has higher tendency to donate e^- .

14 Balancing Redox Reactions

Ion-Electron Method (in acidic medium)

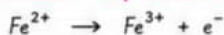
- Write the unbalanced reaction.
- Assign oxidation numbers.
- Write oxidation and reduction half reactions.
- Balance each half reaction (atoms other than H and O).
- Balance O by adding H_2O .
- Balance H by adding H^+ .
- Balance charge by adding e^- .
- Multiply half reactions so that e^- lost = e^- gained.
- Add the half reactions and cancel common species.

Example :

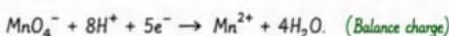
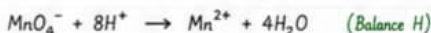
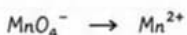
Balance the reaction in acidic medium.



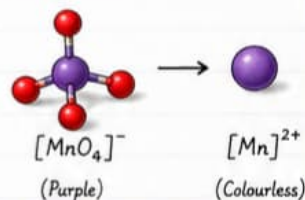
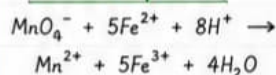
Oxidation Half Reaction :



Reduction Half Reaction :



Balanced Equation :



In basic medium :

After balancing in acidic medium, add OH^- to both sides to neutralise H^+ and then simplify.

15 Some Common Redox Reagents

Reagent	Nature	Medium	Oxidation State Change	Uses
$KMnO_4$	Strong oxidising agent	Acidic or neutral	$Mn^{+7} \rightarrow Mn^{2+}$ (Purple → Colourless)	• Titrations (permanganate titration) • Oxidation of Fe^{2+}, oxalate, nitrite etc.
$K_2Cr_2O_7$	Strong oxidising agent	Acidic	$Cr^{+6} \rightarrow Cr^{3+}$ (Orange → Green)	• Titrations (dichromate titration) • Oxidation of Fe^{2+}, I^-, SO_3^{2-} etc.
H_2O_2	Mild oxidising agent	Acidic, basic or neutral	$O^{-1} \rightarrow O^{-2}$ (Colourless → Colourless)	• Disinfectant • Bleaching agent, hair bleaching
I_2	Mild oxidising agent	Neutral	$I^0 \rightarrow I^-$ (Brown → Colourless)	• Titration (iodometry) • Oxidation of SO_3^{2-}, $S_2O_3^{2-}$ etc.
$SnCl_2$	Strong reducing agent	Acidic	$Sn^{2+} \rightarrow Sn^{4+}$ (Colourless → Colourless)	• Used to reduce Fe^{3+} to Fe^{2+} • In electroplating

Quick Revision

- ✓ Oxidation → loss of e^- / increase in O.N.
- ✓ Reduction → gain of e^- / decrease in O.N.
- ✓ Oxidising agent → gets reduced.
- ✓ Reducing agent → gets oxidised.
- ✓ Redox reactions → oxidation and reduction occur simultaneously.

Memory Tip

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Oxidation Is Loss
Reduction Is Gain

Be Smart! ★

Look for change in oxidation numbers – that tells you which species is oxidised and which is reduced.

